

Seven Seas A-63

Security Audit

September 29th, 2025

Version 1.0.0

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Introduction

This document includes the results of the security audit for Seven Seas's smart contract code as found in the section titled 'Source Code'. The audit was performed by the Macro security team on September 19th, 2025.

The purpose of this audit is to review the source code of certain Seven Seas Solidity contracts, and provide feedback on the design, architecture, and quality of the source code with an emphasis on validating the correctness and security of the software in its entirety.

Disclaimer: While Macro's review is comprehensive and has surfaced some changes that should be made to the source code, this audit should not solely be relied upon for security, as no single audit is guaranteed to catch all possible bugs.

Overall Assessment

The following is an aggregation of issues found by the Macro Audit team:

Severity	Count	Acknowledged	Won't Do	Addressed
Informational	1	-	-	-

Seven Seas was quick to respond to these issues.

Specification

Our understanding of the specification was based on the following sources:

- Discussions with the Seven Seas team.
- Available documentation in the repository.

Source Code

The following source code was reviewed during the audit:

- **Ethena Minting Decoder (from [PR 458](#))**

Commit Hash: `c5d1f09507404393b5447cbbb391546aa66d431a`

Source Code	SHA256
<code>src/base/DecodersAndSanitizers/Protocols/EthenaMintingDecoderAndSanitizer.sol</code>	<code>4d0138537bf90b174c6510b815e2877ca76533ebee4acfe63bb9e06fafa85f4a</code>
<code>src/interfaces/DecoderCustomTypes.sol</code>	<code>276a7ad5447be532f1ad3cffbe25b9a68c5ea13d926568ab14b12188afac3eda</code>

Note: This document contains an audit solely of the Solidity contracts listed above. Specifically, the audit pertains only to the contracts themselves, and does not pertain to any other programs or scripts, including deployment scripts.

Issue Descriptions and Recommendations

Click on an issue to jump to it, or scroll down to see them all.

- I-1 Delegated signer can bypass decoder validation through alternative execution paths

Security Level Reference

We quantify issues in three parts:

1. The high/medium/low/spec-breaking **impact** of the issue:

- How bad things can get (for a vulnerability)
- The significance of an improvement (for a code quality issue)
- The amount of gas saved (for a gas optimization)

2. The high/medium/low **likelihood** of the issue:

- How likely is the issue to occur (for a vulnerability)

3. The overall critical/high/medium/low **severity** of the issue.

This third part – the severity level – is a summary of how much consideration the client should give to fixing the issue. We assign severity according to the table of guidelines below:

Severity	Description
(C-x) Critical	We recommend the client must fix the issue, no matter what, because not fixing would mean significant funds/assets WILL be lost.
(H-x) High	We recommend the client must address the issue, no matter what, because not fixing would be very bad, or some funds/assets will be lost, or the code's behavior is against the provided spec.
(M-x) Medium	We recommend the client to seriously consider fixing the issue, as the implications of not fixing the issue are severe enough to impact the project significantly, albeit not in an existential manner.
(L-x) Low	The risk is small, unlikely, or may not be relevant to the project in a meaningful way. Whether or not the project wants to develop a fix is up to the goals and needs of the project.
(Q-x) Code Quality	The issue identified does not pose any obvious risk, but fixing could improve overall code quality, on-chain composability, developer ergonomics, or even certain aspects of protocol design.
(I-x) Informational	Warnings and things to keep in mind when operating the protocol. No immediate action required.
(G-x) Gas Optimizations	The presented optimization suggestion would save an amount of gas significant enough, in our opinion, to be worth the development cost of implementing it.

Issue Details

I-1

Delegated signer can bypass decoder validation through alternative execution paths

TOPIC	IMPACT
Trust Model	Informational *

The `EthenaMintingDecoderAndSanitizer` validates order parameters when the vault calls Ethena's `mint()` or `redeem()` functions through the `ManagerWithMerkleVerification`. However, once a vault sets a delegated signer, that signer can create and submit orders through alternative execution paths that bypass all decoder protections.

According to Ethena's source code, an order signed by the delegated EOA is valid for the vault, regardless of:

- Who submits the transaction onchain
- What the `order.beneficiary` address is
- What the `order.collateral_asset` is

According to Ethena's API documentation, "orders must be submitted by the benefactor address." This means a compromised delegated EOA cannot directly submit orders to Ethena's API - only the vault (benefactor) can submit orders through the API.

However, Ethena's onchain `mint()` and `redeem()` functions are permissioned with `onlyRole(MINTER_ROLE)` . If there are multiple MINTER_ROLE holders (e.g., the vault itself and Ethena's hot wallets), a compromised delegated EOA could:

1. Sign an order with arbitrary parameters (e.g., `beneficiary = attackerAddress`)
2. Submit this signed order to another MINTER_ROLE holder
3. If that MINTER_ROLE holder executes it onchain, the vault's decoder is bypassed

This would require either:

- Tricking Ethena's systems into executing a malicious order (e.g., through social engineering or exploiting their validation logic)
- Finding another compromised/malicious MINTER_ROLE holder

Trust Model

According to the development team:

- Their vaults will be whitelisted for MINTER_ROLE
- The delegated signer will be their own trusted EOA
- There is no other way to mint through Ethena

Given the above trust assumptions, this design is acceptable. The delegated EOA is explicitly trusted, so its ability to bypass decoder validation is a known and accepted risk of the system.

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The scope of this report and review is limited to a review of only the code presented by the Seven Seas team and only the source code Macro notes as being within the scope of Macro's review within this report. This report does not include an audit of the deployment scripts used to deploy the Solidity contracts in the repository corresponding to this audit. Specifically, for the avoidance of doubt, this report does not constitute investment advice, is not intended to be relied upon as investment advice, is not an endorsement of this project or team, and it is not a guarantee as to the absolute security of the project. In this report you may through hypertext or other computer links, gain access to websites operated by persons other than Macro. Such hyperlinks are provided for your reference and convenience only, and are the exclusive responsibility of such websites' owners. You agree that Macro is not responsible for the content or operation of such websites, and that Macro shall have no liability to your or any other person or entity for the use of third party websites. Macro assumes no responsibility for the use of third party software and shall have no liability whatsoever to any person or entity for the accuracy or completeness of any outcome generated by such software.